

Global and Regional Trends in Herpes Simplex Virus Type 2 Infection and Genital Ulcer Disease

This repository contains MATLAB code and input files used for the manuscript:

Global and Regional Trends in Herpes Simplex Virus Type 2 Infection and Genital Ulcer Disease, 1980–2050: A Mathematical Modeling Study

The code implements a deterministic, compartmental HSV-2 transmission model to estimate historical, current, and projected trends in HSV-2 infection and HSV-2-attributable genital ulcer disease (GUD) globally and across World Health Organization (WHO) regions.

Repository contents

The repository is organized into two main code archives/folders.

1. `Demographic Models Codes`

This folder contains the demographic model codes used to generate sex- and region-specific demographic inputs for the HSV-2 transmission model. Codes are organized by WHO region and sex, including AFRO, AMRO without the United States, EMRO, EURO, SEARO, WPRO, and the United States.

Each region/sex folder includes the relevant MATLAB scripts, input files, and precomputed outputs for demographic fitting and simulation. These outputs can be used directly by the HSV-2 model scripts.

2. `HSV-2 Model Codes`

This folder contains the HSV-2 transmission model, fitting scripts, input datasets, model outputs, and figure files.

Main MATLAB scripts include:

- `HSV2model\_Fun.m`: HSV-2 transmission model ODE system
- `mainHSV2\_fitting.m`: objective/cost function used for model fitting
- `fitting\_prevalence.m`: calibration of the model to HSV-2 prevalence data
- `simulation\_fitting.m`: visualization of model fit to prevalence data
- `simulation\_fitting\_Region\_Global.m`: simulation of regional and global HSV-2 infection and GUD outcomes
- `MainTextOutcomes.m`: generation of main text figures
- `AppendixOutcomes.m`: generation of supplementary figures
- `DataExtraction.m`: extraction and formatting of prevalence calibration data
- `CI\_Sampling\_Triangular.m`: uncertainty interval sampling utility
- `TransitionRates.m`: transition-rate utility

The folder also includes input and output files in `.xlsx` and `.mat` formats, together with main text and supplementary figure files in `.fig`, `.pdf`, and `.tiff` formats.

Main outputs

The code generates estimates for:

- HSV-2 prevalence
- HSV-2 incidence rate
- Annual number of new HSV-2 infections
- Number of individuals experiencing at least one HSV-2-attributable GUD episode within a year
- Annual number of person-days with HSV-2-attributable GUD

Outputs are produced globally and by WHO region, sex, age group, and calendar year, depending on the script used.

Software requirements

The code was developed and run using:

- MATLAB R2019a

The scripts use standard MATLAB functions, including ODE solvers and optimization routines. Some scripts read and write Excel files and therefore require spreadsheet support in MATLAB.

How to run the code

Step 1. Unzip the repository files

Unzip the main repository and then unzip the two included archives:

- `Demographic Models Codes.zip`
- `HSV-2 Model Codes.zip`

Keep the directory structure unchanged.

Step 2. Generate or inspect demographic inputs

Demographic model codes are provided separately by WHO region and sex. To rerun the demographic model for a specific region and sex, open the corresponding folder and run the relevant main script, for example:

```
MainAFRO
```

for the African Region, or the analogous `Main<REGION>.m` script for other regions.

Precomputed demographic output files are also included and can be used directly by the HSV-2 model scripts.

Step 3. Run HSV-2 model simulations

From within the `HSV-2 Model Codes` folder, run:

```
simulation_fitting_Region_Global
```

This script simulates the regional and global HSV-2 transmission model and generates epidemiological and GUD outcome estimates.

Step 4. Generate figures

To generate main text figures, run:

```
MainTextOutcomes
```

To generate supplementary figures, run:

```
AppendixOutcomes
```

The repository also includes pre-generated figure files under:

- `Figures (Main Text)`
- `Figures (Supplementary Material)`

Input data

The repository includes the datasets used for model calibration and simulation, including HSV-2 prevalence data, NHANES data, population size inputs, and precomputed fitted model outputs. Key files include:

- `HSV-2Data\_Age.xlsx`
- `NHANES\_Female.xlsx`
- `NHANES\_Male.xlsx`
- `NHANES\_Female\_All.xlsx`
- `NHANES\_Male\_All.xlsx`
- `Femalepopulationsize.xlsx`
- `MalePopulationsize.xlsx`
- `Prevalence\_Result\_fit.mat`
- `Prevalence\_Result\_fit\_Lower.mat`
- `Prevalence\_Result\_fit\_Upper.mat`

Figure files

Main text figures are provided in:

```
HSV-2 Model Codes/Figures (Main Text)
```

Supplementary figures are provided in:

```
HSV-2 Model Codes/Figures (Supplementary Material)
```

Figures are available in accepted publication formats including `.pdf` and `.tiff`, with MATLAB `.fig` files also included where applicable.

Reproducibility statement

The MATLAB code and input files provided in this repository can reproduce the analyses and figures presented in the article. Precomputed fitted parameter files and model outputs are included to facilitate reproducibility without requiring all calibration steps to be rerun.

Access restrictions

There are no restrictions on access to the code provided in this repository.

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